

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			[An acceleration that applies to] motion on a circular [path] (1) [is directed] towards the centre [of the circle] (1)	2			2		
	(b)	(i)		Period, $T = 24 \times 60 \times 60 = 86400 \text{ s}$ (1) $\omega = \frac{2\pi}{T} = \frac{2\pi}{86400} \text{ (1) [= } 7.27 \times 10^{-5} \text{ rad s}^{-1} \text{]}$		2		2	2	
		(ii)		$a = \omega^2 R_E = (7.27 \times 10^{-5})^2 (6370 \times 10^3) \text{ (1) ecf}$ $= 0.0337 \text{ [m s}^{-2} \text{] (1)}$ [alternatively could use $a = \frac{v^2}{r}$] Accept $0.0312 \text{ [m s}^{-2} \text{]}$ if 7×10^{-5} used	1	1		2	2	
		(iii)	I	$v = \omega r = 7.27 \times 10^{-5} \times 6370 \times 10^3 \text{ (1)}$ $= 463 \text{ [m s}^{-1} \text{] (1)}$ Accept $446 \text{ [m s}^{-1} \text{]}$ if 7×10^{-5} used	1	1		2	2	
			II	Speed is irrelevant / velocity changes slowly / velocity can't be felt (1) {Acceleration / force} is small (1) Don't accept acceleration = 0			2	2		
				Question 2 total	4	4	2	10	6	0